MIL-C-17223A
27 APRIL 1960
SUPERSEDING
MIL-C-17223(SHIPS)
27 JUNE 1952

MILITARY SPECIFICATION

CLEANER, WELDING AND CUTTING TIP

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers tip cleaners consisting of broach-like wires for removing slag or other foreign material restricting or clogging the gas orifices of oxy-fuel gas cutting and welding tips. The tip cleaners specified herein are of the type made of flexible stainless steel wire having a ridged portion near one end. The carbon steel twist drill type is not included.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-T-97 — Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

MIL-W-17481 — Wire, Steel, Corrosion-Resisting.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Material. Tip cleaners shall be made of composition 304 corrosion-resisting steel wire conforming to condition A or B of Specification MIL-W-17481. Cleaning ridges, as specified in 3.3, shall be formed cold.

3.2 Length. Each tip cleaner shall be long enough to ensure that the cleaning ridges will sweep the full length of the sized portion of the gas orifices in any cutting or welding tip having an orifice of the size for which the cleaner is intended. No tip cleaner shall exceed 3⅛ inches in length.

3.3 Ridged cleaning portion. The cleaning portion shall consist of a series of uniformaly spaced, cold-formed, annular or helical ridges of uniform and smooth contour. The tops of the ridges, when examined under low (30 X) magnification shall appear sufficiently smooth so that they cannot cut or scratch the orifice walls. At the same magnification, the valleys between ridges shall appear smooth and free from sharp corners. The outside diameter of the cleaning ridges shall be as specified in table I, dimension Y; and the axial distance between cleaning ridges (pitch) shall be as specified in table I. The ridged cleaning portion shall be ½ to 1 inch long, and may be continuous or in separate sections. If separate, the sum of the lengths shall be ½ to 1 inch.

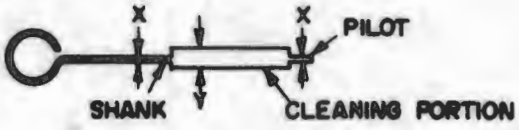


FIGURE 1. *Tip cleaner.*

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TABLE I. *Dimensions.*

Tip cleaner designation	Gas orifice cleaning range		Dimensions			
	Drill numbers	Orifice size (inch)	X [1]	Y [1]	Pitch (max.)	
					Annular	Helical
					Inch	Inch
A	75	0.0210	0.019	0.0210	0.015	0.025
	74	.0225				
B	73	.0240	.023	.0240	.015	.035
	72	.0250				
	71	.0260				
C	70	.0280	.025	.0280	.025	.035
	69	.02925				
	68	.0310				
D	67	.0320	.029	.0320	.025	.040
	66	.0330				
	65	.0350				
E	64	.0360	.033	.0360	.025	.040
	63	.0370				
	62	.0380				
F	61	.0390	.036	.0390	.025	.040
	60	.0400				
G	59	.0410	.038	.0410	.025	.040
	58	.0420				
H	57	.0430	.040	.0430	.025	.040
J	56	.0465	.043	.0465	.025	.040
K	55	.0520	.048	.0520	.035	.050
	54	.0550				
L	53	.0595	.055	.0595	.035	.050
	52	.0635				
M	51	.0670	.063	.0670	.035	.050
	50	.0700				
	49	.0730				
N	48	.0760	.071	.0760	.035	.050
	47	.0785				
O	46	.0810	.076	.0810	.035	.050
	45	.0820				
P	44	.0860	.081	.0860	.035	.050
	43	.0890				
Q	42	.0935	.087	.0935	.035	.050
	41	.0960				
R	40	.0980	.093	.0980	.035	.050
	39	.0995				
	38	.1015				
S	37	.1040	.099	.1040	.035	.050
	36	.1065				
T	35	.1100	.105	.1100	.035	.050
	34	.1110				
	33	.1130				
U	32	.1160	.111	.1160	.035	.050
	31	.1200				
V	30	.1285	.121	.1280	.035	.050

[1] See figure 1 and table II.

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TABLE II. *Tolerances.*

Dimension	Plus inch	Minus inch	Cleaner designation
X	0.000	0.002	A through M
	.000	.003	N through Q
	.000	.004	R through V
Y	.0005	.0025	A through O
	.0005	.0035	P through V

3.4 Pilot end and shank. The pilot end of each tip cleaner shall be ¼ to ½-inch long and the tip shall be rounded or chamfered and free of burrs or sharp corners. The diameter of the pilot and shank shall be as specified in table I, dimension X.

3.5 Sets. Unless otherwise specified tip cleaners shall be furnished in sets consisting of 21 tip cleaners in one case (see 6.2). One of each of the sizes specified in table I shall be included, and sets shall be assembled in consecutive order as specified in table I.

3.6 Cases.

3.6.1 *Material.* Cases shall be made of aluminum-manganese (3S) alloy sheet of suitable thickness. Other parts of case shall be made of a suitable corrosion-resistant metal.

3.6.2 *Size.* Cases shall be compact. Total volume shall not exceed 4 cubic inches.

3.6.3 *Construction.* Cleaners shall be securely fastened inside the case in such manner that easy replacement of individual cleaners is possible. Covers shall be capable of being easily opened and closed without becoming detached. When closed they shall stay closed under normal handling. Cases may be either single or multiple-compartmented; but shall be constructed so that any individual cleaner can be swung out through at least 180 degrees. Individual cleaners shall be conveniently usable without necessity for detaching from the case, and with no interference from the case or other cleaners.

3.6.4 *Marking.* The outside of each case shall be clearly and indelibly marked with the name of the manufacturer or supplier, or with a neat distinctive trademark by which the manufacturer or supplier can be readily identified. The tip designations and corresponding drill size numbers, as specified in table I, shall also be clearly stamped in the metal of the case on the outside.

3.7 Workmanship. The torch-tip cleaner shall be free from all imperfections which may impair the serviceability. All parts of the case shall be smooth and free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Lot. All sets presented at one time shall be considered a lot for purposes of acceptance inspection.

4.3 Sampling for visual and dimensional examination. A random sample of sets shall be selected at inspection level II; the acceptable quality level (AQL) shall be 2.5 percent defective, from each lot of material offered for examination of visual and dimensional characteristics with lot acceptance based on the single sampling inspection requirements in accordance with Standard MIL–STD–105.

4.4 Visual and dimensional examination.

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Each of the sample sets selected in accordance with 4.3 shall be visually and dimensionally examined by the inspector to verify compliance with this specification. Any set in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective sets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for examination provided the contractor has removed all nonconforming sets. Another sample of cleaners shall be selected and examined to verify conformance to this specification.

4.5 **Inspection of preparation for delivery.** Sample items and packages shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Tip cleaners shall be packaged level A or C, as specified (see 6.2).

5.1.1 *Level A.* Tip cleaners shall be packaged in accordance with method III of Specification MIL-P-116. Quantity per unit package shall be as specified (see 6.2).

5.1.2 *Level C.* Tip cleaners shall be packaged in accordance with the manufacturer's commercial practice.

5.2 **Packing.** Tip cleaners shall be packed level A, B or C as specified (see 6.2).

5.2.1 *Quantity per shipping container.* The quantity per shipping container shall be as specified (see 6.2). In the absence of specified quantities, the weight limitations shown in 5.2.2 and 5.2.3 shall apply.

5.2.2 *Level A.* Tip cleaners shall be packed in wood-cleated fiberboard, nailed wood, fiber, wirebound wood or wood cleated-plywood boxes conforming to Specification PPP-B-591 (overseas type), PPP-B-621 (class 2), PPP-B-636 (class 3, compliance symbol V3c or V3s), PPP-B-585 (class 2 or 3), or PPP-B-601 (overseas type), respectively, at the option of the contractor. Gross weight of wood type boxes shall not exceed the weight limitations of the box specification. Box closures and strapping shall conform to the box specification and the appendix thereto. Boxes shall be provided with case liners conforming to Specification MIL-L-10547. Case liners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. In lieu of steel straps fiber boxes may be banded with tape conforming to type IV of Specification PPP-T-97 and the appendix thereto.

5.2.3 *Level B.* Tip cleaners shall be packed in wood-cleated fiberboard, nailed wood, wirebound wood, wood cleated-plywood, wood cleated veneer paper overlaid or fiber boxes conforming to Specification PPP-B-591 (domestic type) PPP-B-621 (class 1), PPP-B-585 (class 1), PPP-B-601 (domestic type), MIL-B-10877 (domestic type) or PPP-B-636 (class 2), respectively, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto, except strapping of class 2, PPP-B-636 fiber box shall not be required. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.4 *Level C.* Tip cleaners shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. Tip cleaners are intended for use in removing carbon deposits in orifices of welding or cutting tips.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Tip cleaner sets, if different (see 3.5).

(c) Selection of the level of packaging and the level of packing (see 5.1 and 5.2).

(d) Quantity per unit package (see 5.1.1).

(e) Quantity per shipping container (see 5.2.1).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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